

Differentiation Rules - Solutions

LLE – Mathematics and Statistics

Power Rule

Given the function $y = ax^n$, the derivative is:

$$\frac{dy}{dx} = nax^{n-1}$$

Solutions for differentiation using the power rule:

1. **Question:** $y = 5x^4$

Solution: Apply the power rule directly.

$$\frac{dy}{dx} = 4 \times 5x^{4-1} = 20x^3$$

2. **Question:** $y = 7x^2$

Solution: Apply the power rule directly.

$$\frac{dy}{dx} = 2 \times 7x^{2-1} = 14x$$

3. **Question:** $y = \frac{2}{3}x^6$

Solution: Apply the power rule directly.

$$\frac{dy}{dx} = 6 \times \frac{2}{3}x^{6-1} = 4x^5$$

Solutions for differentiation using indices facts and the power rule:

4. **Question:** $y = \frac{3x^3}{x^5}$

Solution: First, simplify the expression using index laws: $\frac{x^a}{x^b} = x^{a-b}$.

$$y = 3x^{3-5} = 3x^{-2}$$

Now, apply the power rule.

$$\frac{dy}{dx} = -2 \times 3x^{-2-1} = -6x^{-3} = -\frac{6}{x^3}$$

5. **Question:** $y = \frac{10}{x^2}$

Solution: First, rewrite the expression using index laws: $\frac{1}{x^n} = x^{-n}$.

$$y = 10x^{-2}$$

Now, apply the power rule.

$$\frac{dy}{dx} = -2 \times 10x^{-2-1} = -20x^{-3} = -\frac{20}{x^3}$$

6. **Question:** $y = 2x^{\frac{2}{3}}$

Solution: Apply the power rule directly.

$$\frac{dy}{dx} = \frac{2}{3} \times 2x^{\frac{2}{3}-1} = \frac{4}{3}x^{-\frac{1}{3}} = \frac{4}{3\sqrt[3]{x}}$$

7. **Question:** $y = 4\sqrt{x}$

Solution: First, rewrite the expression using index laws: $\sqrt[n]{x} = x^{\frac{1}{n}}$.

$$y = 4x^{\frac{1}{2}}$$

Now, apply the power rule.

$$\frac{dy}{dx} = \frac{1}{2} \times 4x^{\frac{1}{2}-1} = 2x^{-\frac{1}{2}} = \frac{2}{\sqrt{x}}$$

8. **Question:** $y = \frac{2}{3x^2}$

Solution: First, rewrite the expression using index laws: $\frac{1}{x^n} = x^{-n}$.

$$y = \frac{2}{3}x^{-2}$$

Now, apply the power rule.

$$\frac{dy}{dx} = -2 \times \frac{2}{3}x^{-2-1} = -\frac{4}{3}x^{-3} = -\frac{4}{3x^3}$$

9. **Question:** $y = \frac{10}{x}$

Solution: First, rewrite the expression using index laws: $\frac{1}{x} = x^{-1}$.

$$y = 10x^{-1}$$

Now, apply the power rule.

$$\frac{dy}{dx} = -1 \times 10x^{-1-1} = -10x^{-2} = -\frac{10}{x^2}$$

Solutions for differentiating combined functions:

10. **Question:** $y = 5x^2 + 4x + 3$

Solution: Differentiate each term separately using the power rule. The derivative of $4x$ is 4, and the derivative of a constant 3 is 0.

$$\frac{dy}{dx} = 10x + 4$$

11. **Question:** $y = x^4 - 2x^{-5} + \frac{1}{x}$

Solution: Rewrite $\frac{1}{x}$ as x^{-1} , then differentiate each term.

$$y = x^4 - 2x^{-5} + x^{-1}$$

$$\begin{aligned}\frac{dy}{dx} &= 4x^{4-1} - 5 \times (-2)x^{-5-1} + (-1)x^{-1-1} \\ \frac{dy}{dx} &= 4x^3 + 10x^{-6} - x^{-2} = 4x^3 + \frac{10}{x^6} - \frac{1}{x^2}\end{aligned}$$

12. **Question:** $y = 1 + 2\sqrt{x} + \frac{2}{5x} + \frac{3}{4x^2} + \frac{5}{2x^3}$

Solution: Rewrite all terms with fractional or negative exponents first.

$$y = 1 + 2x^{\frac{1}{2}} + \frac{2}{5}x^{-1} + \frac{3}{4}x^{-2} + \frac{5}{2}x^{-3}$$

Differentiate each term.

$$\begin{aligned}\frac{dy}{dx} &= 0 + \frac{1}{2} \times 2x^{\frac{1}{2}-1} + (-1) \times \frac{2}{5}x^{-1-1} + (-2) \times \frac{3}{4}x^{-2-1} + (-3) \times \frac{5}{2}x^{-3-1} \\ \frac{dy}{dx} &= x^{-\frac{1}{2}} - \frac{2}{5}x^{-2} - \frac{3}{2}x^{-3} - \frac{15}{2}x^{-4} \\ \frac{dy}{dx} &= \frac{1}{\sqrt{x}} - \frac{2}{5x^2} - \frac{3}{2x^3} - \frac{15}{2x^4}\end{aligned}$$

Exponentials

Given the function $y = ae^x$, the derivative is:

$$\frac{dy}{dx} = ae^x$$

Solution for differentiation of exponentials:

1. **Question:** $y = 5e^x$

Solution: Apply the exponential rule directly.

$$\frac{dy}{dx} = 5e^x$$

Logarithms

Given the function $y = a \ln x$, the derivative is:

$$\frac{dy}{dx} = \frac{a}{x}$$

Solutions for differentiation of logarithms:

1. **Question:** $y = 5 \ln x$

Solution: Apply the logarithm rule directly.

$$\frac{dy}{dx} = \frac{5}{x}$$

2. **Question:** $y = \frac{\ln x}{2} - 3e^x$

Solution: Rewrite as $y = \frac{1}{2} \ln x - 3e^x$, then differentiate each term.

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{1}{x} - 3e^x = \frac{1}{2x} - 3e^x$$

Solutions for differentiation using log laws:

3. **Question:** $y = \ln 4x$

Solution: Use the log law $\ln(AB) = \ln A + \ln B$ to expand first.

$$y = \ln 4 + \ln x$$

Now differentiate. $\ln 4$ is a constant, so its derivative is 0.

$$\frac{dy}{dx} = 0 + \frac{1}{x} = \frac{1}{x}$$

4. **Question:** $y = \ln 0.443x$

Solution: Use the log law $\ln(AB) = \ln A + \ln B$ to expand first.

$$y = \ln 0.443 + \ln x$$

Now differentiate. $\ln 0.443$ is a constant, so its derivative is 0.

$$\frac{dy}{dx} = 0 + \frac{1}{x} = \frac{1}{x}$$

5. **Question:** $y = \ln \frac{2}{x}$

Solution: Use the log law $\ln\left(\frac{A}{B}\right) = \ln A - \ln B$ to expand first.

$$y = \ln 2 - \ln x$$

Now differentiate. $\ln 2$ is a constant, so its derivative is 0.

$$\frac{dy}{dx} = 0 - \frac{1}{x} = -\frac{1}{x}$$

6. **Question:** $y = \ln x^{10}$

Solution: Use the log law $\ln(A^n) = n \ln A$ to simplify first.

$$y = 10 \ln x$$

Now differentiate.

$$\frac{dy}{dx} = 10 \times \frac{1}{x} = \frac{10}{x}$$

7. **Question:** $y = 6 \ln \frac{1}{2}x$

Solution: Use the log law $\ln(AB) = \ln A + \ln B$ to expand first.

$$y = 6 \left(\ln \frac{1}{2} + \ln x \right) = 6 \ln \frac{1}{2} + 6 \ln x$$

Now differentiate. $6 \ln \frac{1}{2}$ is a constant, so its derivative is 0.

$$\frac{dy}{dx} = 0 + 6 \times \frac{1}{x} = \frac{6}{x}$$

Mixed Questions

Solutions for finding the requested derivative:

1. **Question:** $y = 5x^3 + x^2 - 4x + 1 + 3e^x + \ln 2x$ $\frac{dy}{dx} =$

Solution: Differentiate each term using the appropriate rules. For $\ln 2x$, use $\ln(AB) = \ln A + \ln B$ to get $\ln 2 + \ln x$, then differentiate.

$$\frac{dy}{dx} = 15x^2 + 2x - 4 + 3e^x + \frac{1}{x}$$

2. **Question:** $y = 2x + 5\sqrt{x} - \frac{4}{x} + \frac{3}{x^2}$ $\frac{dy}{dx} =$

Solution: Rewrite terms with fractional or negative exponents: $y = 2x + 5x^{\frac{1}{2}} - 4x^{-1} + 3x^{-2}$. Then differentiate each term.

$$\frac{dy}{dx} = 2 + \frac{5}{2}x^{-\frac{1}{2}} + 4x^{-2} - 6x^{-3} = 2 + \frac{5}{2\sqrt{x}} + \frac{4}{x^2} - \frac{6}{x^3}$$

3. **Question:** $y = 5t^2 - 2t^{\frac{3}{4}} + 6 \ln 0.1t$ $\frac{dy}{dt} =$

Solution: Differentiate each term. For $6 \ln 0.1t$, use $\ln(AB) = \ln A + \ln B$ to get $6(\ln 0.1 + \ln t)$, then differentiate.

$$\frac{dy}{dt} = 10t - \frac{3}{2}t^{-\frac{1}{4}} + \frac{6}{t} = 10t - \frac{3}{2\sqrt[4]{t}} + \frac{6}{t}$$

4. **Question:** $P = 6e^n - 4 + \frac{5}{2}n - \frac{6}{\sqrt{n}}$ $\frac{dP}{dn} =$

Solution: Rewrite terms with fractional or negative exponents: $P = 6e^n - 4 + \frac{5}{2}n - 6n^{-\frac{1}{2}}$. Then differentiate each term.

$$\frac{dP}{dn} = 6e^n + \frac{5}{2} + 3n^{-\frac{3}{2}} = 6e^n + \frac{5}{2} + \frac{3}{n\sqrt{n}}$$

5. **Question:** $G = 5s^2(3s - 4)$ $\frac{dG}{ds} =$

Solution: First, expand the expression: $G = 15s^3 - 20s^2$. Then differentiate each term.

$$\frac{dG}{ds} = 45s^2 - 40s$$

Solutions for finding the value of the function and the gradient at given x values:

6. **Question:** $y = 5x^2 - 12x + 2$, $x = 2$

Solution:

Value of the function at $x = 2$:

$$y(2) = 5(2)^2 - 12(2) + 2 = 5(4) - 24 + 2 = 20 - 24 + 2 = -2$$

Derivative:

$$\frac{dy}{dx} = 10x - 12$$

Value of the gradient at $x = 2$:

$$\frac{dy}{dx}(2) = 10(2) - 12 = 20 - 12 = 8$$

Final Answer: Function value: -2 , Gradient: 8

7. **Question:** $y = 4x^3 + x + 10$, $x = -1$

Solution:

Value of the function at $x = -1$:

$$y(-1) = 4(-1)^3 + (-1) + 10 = 4(-1) - 1 + 10 = 5$$

Derivative:

$$\frac{dy}{dx} = 12x^2 + 1$$

Value of the gradient at $x = -1$:

$$\frac{dy}{dx}(-1) = 12(-1)^2 + 1 = 12(1) + 1 = 13$$

Final Answer: Function value: 5 , Gradient: 13

8. **Question:** $y = x^2 + 5x - 1$, $x = -2.5$

Solution:

Value of the function at $x = -2.5$:

$$y(-2.5) = (-2.5)^2 + 5(-2.5) - 1 = 6.25 - 12.5 - 1 = -7.25$$

Derivative:

$$\frac{dy}{dx} = 2x + 5$$

Value of the gradient at $x = -2.5$:

$$\frac{dy}{dx}(-2.5) = 2(-2.5) + 5 = -5 + 5 = 0$$

Final Answer: Function value: -7.25 , Gradient: 0

9. **Question:** $y = 4e^x - 10\sqrt{x}$, $x = 2$

Solution:

Value of the function at $x = 2$:

$$y(2) = 4e^2 - 10\sqrt{2} \approx 4(7.389) - 10(1.414) = 29.556 - 14.14 = 15.416$$

Derivative:

$$y = 4e^x - 10x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = 4e^x - 10\left(\frac{1}{2}\right)x^{-\frac{1}{2}} = 4e^x - \frac{5}{\sqrt{x}}$$

Value of the gradient at $x = 2$:

$$\frac{dy}{dx}(2) = 4e^2 - \frac{5}{\sqrt{2}} \approx 29.556 - \frac{5}{1.414} = 29.556 - 3.536 = 26.02$$

Final Answer: Function value: ≈ 15.416 , Gradient: ≈ 26.02

10. **Question:** $y = 5 - \frac{5}{x} + \frac{5}{x^2}$, $x = 1$

Solution:

Value of the function at $x = 1$:

$$y(1) = 5 - \frac{5}{1} + \frac{5}{1^2} = 5 - 5 + 5 = 5$$

Derivative:

$$y = 5 - 5x^{-1} + 5x^{-2} \Rightarrow \frac{dy}{dx} = 0 - 5(-1)x^{-2} + 5(-2)x^{-3} = 5x^{-2} - 10x^{-3} = \frac{5}{x^2} - \frac{10}{x^3}$$

Value of the gradient at $x = 1$:

$$\frac{dy}{dx}(1) = \frac{5}{1^2} - \frac{10}{1^3} = 5 - 10 = -5$$

Final Answer: Function value: 5 , Gradient: -5

Consumer Optimisation Extension using Partial Differentiation

- Solutions

A consumer aims to maximize their utility from consuming two goods, x and y . Their utility function is given by:

$$U(x, y) = x^{0.8}y^{0.2}$$

The prices of the goods are $P_x = 4$ and $P_y = 2$, and the consumer has an income of $I = 200$.

Part of a solution:

Budget Constraint:

$$4x + 2y = 200$$

Lagrangian:

$$L(x, y, \lambda) = x^{0.8}y^{0.2} - \lambda(200 - 4x - 2y)$$

1. **Question:** The Lagrangian function is provided above.

Solution: No action required for this step, the Lagrangian is given.

First Order Conditions:

Differentiate the Lagrangian with respect to x , y , and λ , and set each derivative to zero. Remember to apply the power rule for differentiation.

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial L}{\partial y} = 0$$

$$\frac{\partial L}{\partial \lambda} = 0$$

2. **Question:** Write down the full first-order conditions, performing the differentiation steps.

Solution:

$$\frac{\partial L}{\partial x} = 0.8x^{-0.2}y^{0.2} - 4\lambda = 0 \Rightarrow 0.8x^{-0.2}y^{0.2} = 4\lambda$$

$$\frac{\partial L}{\partial y} = 0.2x^{0.8}y^{-0.8} - 2\lambda = 0 \Rightarrow 0.2x^{0.8}y^{-0.8} = 2\lambda$$

$$\frac{\partial L}{\partial \lambda} = -(200 - 4x - 2y) = 0 \Rightarrow 200 - 4x - 2y = 0 \Rightarrow 4x + 2y = 200$$

Solving for Optimal Quantities:

3. **Question:** You can rearrange $\frac{\partial L}{\partial x}$ to get: $0.8x^{-0.2}y^{0.2} = 4\lambda$. Now, create a similar rearranged expression for $\frac{\partial L}{\partial y}$.

Solution: Rearranged $\frac{\partial L}{\partial x}$: $0.8x^{-0.2}y^{0.2} = 4\lambda$

Rearranged $\frac{\partial L}{\partial y}$: $0.2x^{0.8}y^{-0.8} = 2\lambda$

Dividing the Equations:

You then divide these two rearranged equations by one another:

$$\frac{0.8x^{-0.2}y^{0.2}}{0.2x^{0.8}y^{-0.8}} = \frac{4\lambda}{2\lambda}$$

4. **Question:** Substitute your rearranged expression for $\frac{\partial L}{\partial y}$ from step 3 into the ratio above.

Solution: Substitute the expressions derived in step 3.

$$\frac{0.8x^{-0.2}y^{0.2}}{0.2x^{0.8}y^{-0.8}} = \frac{4\lambda}{2\lambda}$$

5. **Question:** Simplify the resulting equation to express y in terms of x .

Solution: Simplify the numerical coefficients and combine the x and y terms using

index laws.

$$4x^{-0.2-0.8}y^{0.2-(-0.8)} = 2$$

$$4x^{-1}y^1 = 2$$

$$\frac{4y}{x} = 2$$

$$4y = 2x$$

$$y = \frac{2x}{4} = \frac{1}{2}x$$

6. **Question:** Substitute your equation for y into the budget constraint ($4x + 2y = 200$).

Solution: Substitute $y = \frac{1}{2}x$ into the budget constraint.

$$4x + 2\left(\frac{1}{2}x\right) = 200$$

$$4x + x = 200$$

$$5x = 200$$

7. **Question:** Solve the resulting equation for x .

Solution:

$$x = \frac{200}{5} = 40$$

8. **Question:** Use the optimal x value to find the optimal quantity of y .

Solution: Substitute $x = 40$ into $y = \frac{1}{2}x$.

$$y = \frac{1}{2}(40) = 20$$

Final Answer: Optimal quantities: $x = 40, y = 20$

Extension Question: Utility Maximisation with Logarithmic Functions - Solutions

A consumer aims to maximize their utility from consuming two goods, x and y . Their utility function is given by:

$$U(x, y) = 2 \ln x + 3 \ln y$$

The prices of the goods are $P_x = 5$ and $P_y = 10$, and the consumer has an income of $I = 100$.

Part of a solution:

Budget Constraint:

$$5x + 10y = 100$$

Lagrangian:

$$L(x, y, \lambda) = 2 \ln x + 3 \ln y - \lambda(100 - 5x - 10y)$$

1. **Question:** The Lagrangian function is provided above.

Solution: No action required for this step, the Lagrangian is given.

First Order Conditions:

Differentiate the Lagrangian with respect to x , y , and λ , and set each derivative to zero. Remember to apply the appropriate differentiation rules for logarithmic functions.

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial L}{\partial y} = 0$$

$$\frac{\partial L}{\partial \lambda} = 0$$

2. **Question:** Write down the full first-order conditions, performing the differentiation steps.

Solution:

$$\frac{\partial L}{\partial x} = \frac{2}{x} - 5\lambda = 0 \Rightarrow \frac{2}{x} = 5\lambda$$

$$\frac{\partial L}{\partial y} = \frac{3}{y} - 10\lambda = 0 \Rightarrow \frac{3}{y} = 10\lambda$$

$$\frac{\partial L}{\partial \lambda} = -(100 - 5x - 10y) = 0 \Rightarrow 100 - 5x - 10y = 0 \Rightarrow 5x + 10y = 100$$

Solving for Optimal Quantities:

3. **Question:** You can rearrange $\frac{\partial L}{\partial x}$ to get: $\frac{2}{x} = 5\lambda$. Now, create a similar rearranged expression for $\frac{\partial L}{\partial y}$.

Solution: Rearranged $\frac{\partial L}{\partial x}$: $\frac{2}{x} = 5\lambda$

Rearranged $\frac{\partial L}{\partial y}$: $\frac{3}{y} = 10\lambda$

Dividing the Equations:

You then divide these two rearranged equations by one another:

$$\frac{2x^{-1}}{3y^{-1}} = \frac{5\lambda}{10\lambda}$$

4. **Question:** Substitute your rearranged expression for $\frac{\partial L}{\partial y}$ from step 3 into the ratio above.

Solution: Substitute the expressions derived in step 3.

$$\frac{2x^{-1}}{3y^{-1}} = \frac{5\lambda}{10\lambda}$$

5. **Question:** Simplify the resulting equation to express y in terms of x .

Solution: Simplify the numerical coefficients and rearrange using index laws.

$$\begin{aligned}\frac{2}{3}x^{-1}y^1 &= \frac{1}{2} \\ \frac{2y}{3x} &= \frac{1}{2} \\ 4y &= 3x \\ y &= \frac{3}{4}x\end{aligned}$$

6. **Question:** Substitute your equation for y into the budget constraint ($5x + 10y = 100$).

Solution: Substitute $y = \frac{3}{4}x$ into the budget constraint.

$$\begin{aligned}5x + 10\left(\frac{3}{4}x\right) &= 100 \\ 5x + \frac{30}{4}x &= 100 \\ 5x + 7.5x &= 100 \\ 12.5x &= 100\end{aligned}$$

7. **Question:** Solve the resulting equation for x .

Solution:

$$x = \frac{100}{12.5} = 8$$

8. **Question:** Use the optimal x value to find the optimal quantity of y .

Solution: Substitute $x = 8$ into $y = \frac{3}{4}x$.

$$y = \frac{3}{4}(8) = 6$$

Final Answer: Optimal quantities: $x = 8, y = 6$